

We sincerely thank all reviewers for valuable comments.

Innovation. We apologize for the unclear statement of our core innovation in Lines 120–124. In this paper, **we rethink the mechanism of social media communication and propose a dual-factor model**, where social media popularity is governed by two independent factors: (1) **content quality** — how engaging the content itself is; (2) **distribution mechanism** — how many users the platform exposes the content to, determined by social attributes (author, category, language, etc.). That is to say, **social attributes should be regarded as proxy variables for modeling platform’s distribution mechanism**, rather than auxiliary variables for measuring content similarity (RAGTrans, SKAPP). Therefore, we can naturally obtain a popularity model as:

$$pop = \text{Factor}(\text{content}) \times \text{Factor}(\text{social attributes}) \quad (1)$$

Architecture. Following this theory, we design the RE-Rag, the first framework to explicitly model both factors within retrieval-augmented architecture. Specifically, in the retrieval stage, image and text embeddings are adopted to measure content similarity, while **the social-attribute-based rarity score** is introduced to measure the potential exposure dominated by platform’s distribution mechanism. In the prediction stage, **we use the relation map constructed by social attributes** to calculate the **distribution mechanism similarity**, which then serves as an attention to guide the subsequent fusion.

Comparison to SKAPP and RAGTrans. As shown in Fig. 1, both SKAPP and RAGTrans utilize social attributes, but their modeling strategies differ fundamentally from RE-Rag. SKAPP treats attributes as auxiliary cues to filter or enhance similar samples. RAGTrans first clusters content into virtual attributes, then concatenates those with social attributes for retrieval. Although RAGTrans introduces social-attribute-based hypergraph in prediction stage, this only provides a structural feature aggregation pattern, which cannot model the distribution mechanism. As a result, our RE-RAG gets superior performance in both retrieval stage and prediction stage. From Tab. 5, **using our retrieval block could improve the performance of SKAPP (MAE:0.649→0.567) and RAGTrans (MAE:0.654→0.582). Besides, the relation-map-based attention fusion brings 0.124 MAE reduction compared to the structural feature aggregation in RAGTrans.**

Experimental Discussion. As shown in Tables 1 and 2, the limited improvement on ICIP is mainly due to its weaker distribution-related attributes (ispro, HasStats) and smaller data scale which weakens the modeling of platform distribution mechanisms and leads to similar performance across methods. In contrast, SMPD and SMTPD provide **sufficient data and attributes highly correlated with distribution mechanisms**, allowing RE-Rag to better exploit attribute-guided relation modeling. This is particularly evident on SMTPD, This indicates that relying solely **content-only similarity insufficient for capturing real propagation differences.**

Relation Modeling. To clarify, the strength of relation is **continuous and learnable**, with discrete social attributes acting only as **structured priors rather than fixed final weights**. This design is necessary because **semantic similarity**

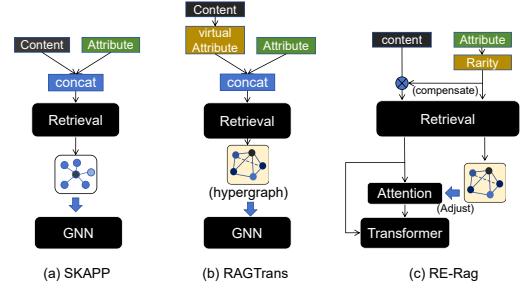


Fig. 1: Differences among various methods

alone struggles to capture explicit categorical boundaries and differences in audience pools (e.g., “Games” and “Celebrities” may be semantically close but have distinct propagation paths). Therefore, We define **three relation priors (0/1/2)** for three retrieval scenarios: 0 for content-similar but distribution-different, 1 for partial social association that provides cross-community references, and 2 for high alignment in both content and propagation paths. Rather than fixed aggregation, RE-Rag uses MLP to transform attribute graphs into continuous attention weights, which allows it to consistently outperform strong RAG baselines under identical input (Tab. 5).

Generalization. We acknowledge that model performance may be affected under extreme social data scarcity. Nevertheless, **RAG naturally improves generalization by leveraging an external retrieval database** that can be dynamically updated with newly observed samples. Instead of memorizing specific users or fixed historical relations, RE-Rag transfers popularity patterns from relevant retrieved instances to the target sample. Thus, the model can better adapt to evolving platform distributions, even when partial social attributes are unavailable.

Experimental Supplement. (1) Attention Weight Visualization. Fig. 2 visualizes the attention weights of 40 query samples over their Top-20 retrieved samples, with brighter colors indicating larger weights. Compared with the model without relation modeling in Fig. 2(a), the model with relation modeling in Fig. 2(b) shows a more continuous and discriminative attention distribution, rather than concentrating mainly on a few top-ranked samples. (2) **Contribution of Social Attributes.** As shown in Supplementary Table 1, all attributes improve performance to varying degrees. “Author” contributes most, followed by “Category,” suggesting that more discriminative attributes yield greater information gains.

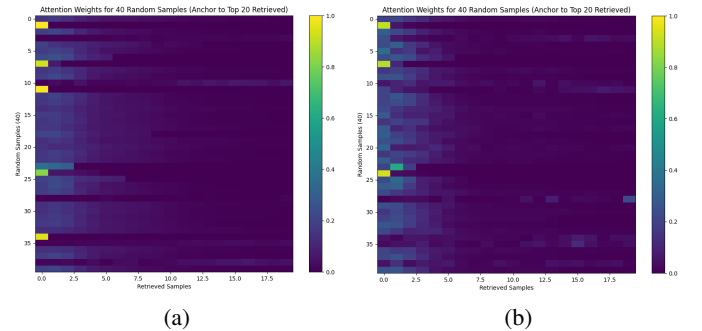


Fig. 2: Attention Weight Visualization